

PASTA worksheet

Stages	Sneaker company
I. Define business and security objectives	• Connect buyers and sellers through user accounts, direct messaging, and seller ratings. • Protect user privacy and handle user information responsibly. • Process sales quickly and clearly, provide multiple payment options, and meet legal requirements for payment handling.
II. Define the technical scope	• SQL – use prepared or parameterized statements for database queries. • API – enforce authentication and authorization when applications communicate. • PKI – use secure encryption to protect information exchanged between the app and users.
III. Decompose application	The data-flow diagram shows a product search process that interacts with the user database . Users search for sneakers for sale, and the application provides listings of current inventory.
IV. Threat analysis	• SQL injection • Session hijacking These are both identified as attack paths against user data in the provided attack tree.
V. Vulnerability analysis	• Lack of prepared statements • Weak login credentials <i>These vulnerabilities are identified in the supplied attack tree as</i>

	<i>weaknesses that can support the listed attacks.</i>
VI. Attack modeling	User data → SQL injection → Lack of prepared statements User data → Session hijacking → Weak login credentials
VII. Risk analysis and impact	• Use prepared statements and validate user input before processing SQL queries. • Require strong login credentials and use secure session management. • Use PKI/TLS encryption to protect data while it is being transmitted. • Encrypt sensitive data, such as payment information, and apply appropriate access controls.
